Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PH611 - NUCLEAR AND RADIATION PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. What are magic numbers?
2. Write semi empirical mass formula.
3. Define nuclear isomerism.
4. What are gamma rays?
5. What is meant by particle accelerator?
6. Give the principle of Synchrocyclotron.
7. What is nuclear fission?
8. What is a chain reaction?
9. Explain about strange particles.
10. Write short note on Strong interaction.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Explain the liquid drop model of the nucleus.

(or)

b) Discuss the meson theory of nuclear forces.

12. a) State and explain Geiger – Nuttal law.

(or)

b) Explain the theory of alpha decay in detail.

13. a) Describe the construction and working of an Ionization chamber.

(or)

b) Describe the construction and action of a GM counter.

14. a) Write short notes on pressurized heavy water reactor.

(or)

b) Briefly explain about fast breeder reactor.

15. a) Describe the basic ideas of Quarks.

(or)

b) What are leptons? Name the different kinds of leptons. Give the general properties of leptons.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Explain the evidences for the shell model of the nucleus.

17. Explain the neutrino theory of beta decay.

18. Describe the construction and working of a Betatron.

19. Deduce the four factor formula from the nuclear chain reaction.

20. Describe the different kinds of elementary particles and explain their properties.

****